

PVsyst - Simulation report

Grid-Connected System

Project: Fazouane

Variant: Fazouane

Sheds on ground

System power: 3200 Wp

Fazouane - Morocco



PVsyst V8.1.6

VC0, Simulation date:
30/09/26 22:20
with V8.1.6

Project summary

Geographical Site

Fazouane
Morocco

Situation

Latitude 34.9100 °(N)
Longitude -2.2000 °(W)
Altitude 450 m
Time zone UTC

Project settings

Albedo 0.20

Weather data

Fazouane
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Orientation #1

Fixed plane

Tilt/Azimuth 30 / 0 °

System information

PV Array

Nb. of modules
Pnom total

Sheds on ground

Near Shadings

Linear shadings : Fast (table)

8 units
3200 Wp

Inverters

Nb. of units
Total power
Pnom ratio

User's needs

Unlimited load (grid)

1 unit
3 kWac
1.07

Results summary

Simulation step Hourly

Produced Energy 5605.6 kWh/year Specific production 1752 kWh/kWp/year Perf. Ratio PR 83.30 %

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General parameters**Grid-Connected System****Sheds on ground****Simulation step**

Hourly

Orientation #1**Fixed plane**

Tilt/Azimuth 30 / 0 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics**PV module**

Manufacturer Generic
Model Mono 400 Wp 72 cells
(Original PVsyst database)
Unit Nom. Power 400 Wp
Number of PV modules 8 units
Nominal (STC) 3200 Wp
Modules 1 unit x 8 In series

At operating cond. (50°C)

Pmpp 2901 Wp
U mpp 276 V
I mpp 11 A

Total PV power

Nominal (STC) 3.20 kWp
Total 8 modules
Module area 17.9 m²
Cell area 15.9 m²

Inverter

Manufacturer Generic
Model 3 kWac inverter
(Original PVsyst database)
Unit Nom. Power 3.00 kWac
Number of inverters 1 unit
Total power 3.0 kWac
Operating voltage 125-440 V
Pnom ratio (DC:AC) 1.07

Total inverter power

Total power 3 kWac
Number of inverters 1 unit
Pnom ratio 1.07

Array losses**Thermal Loss factor**

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 440 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.38 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

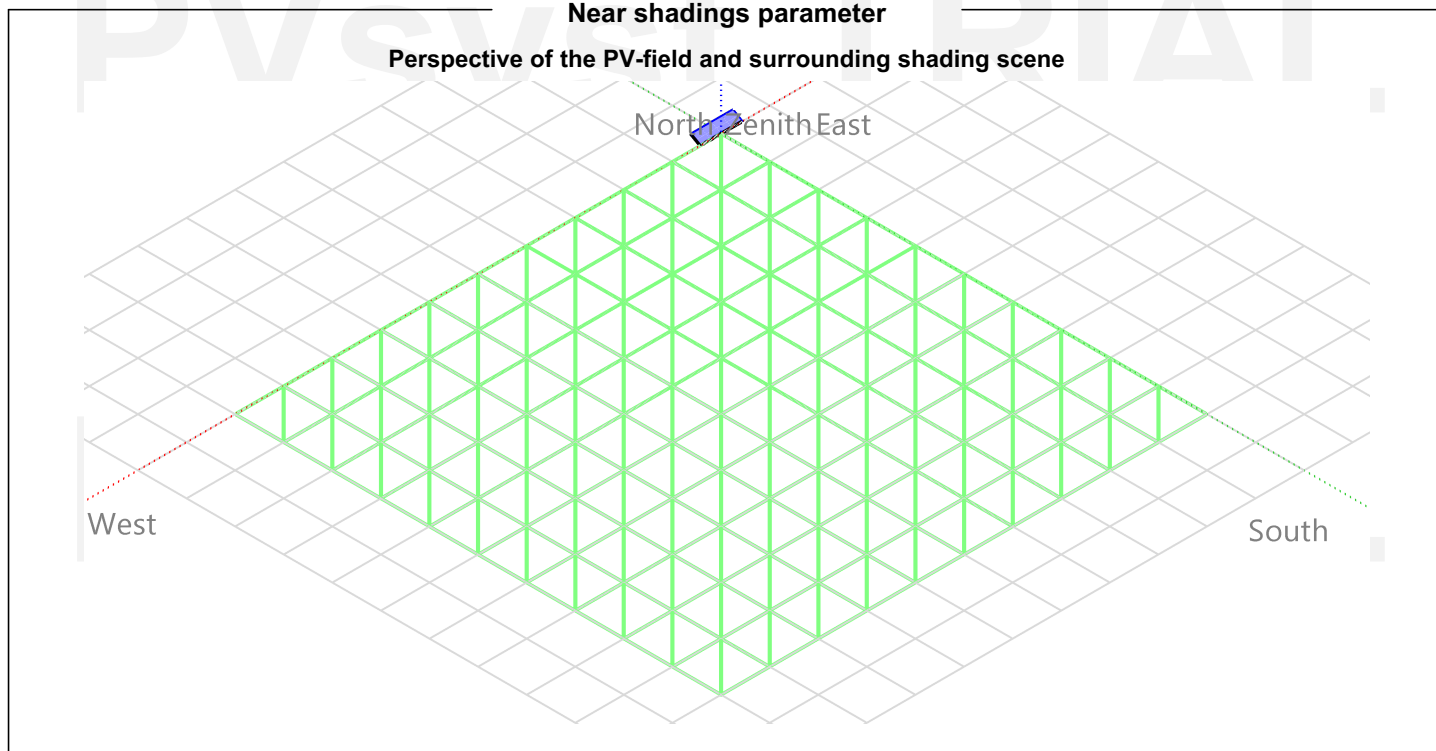


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Near shadings parameter

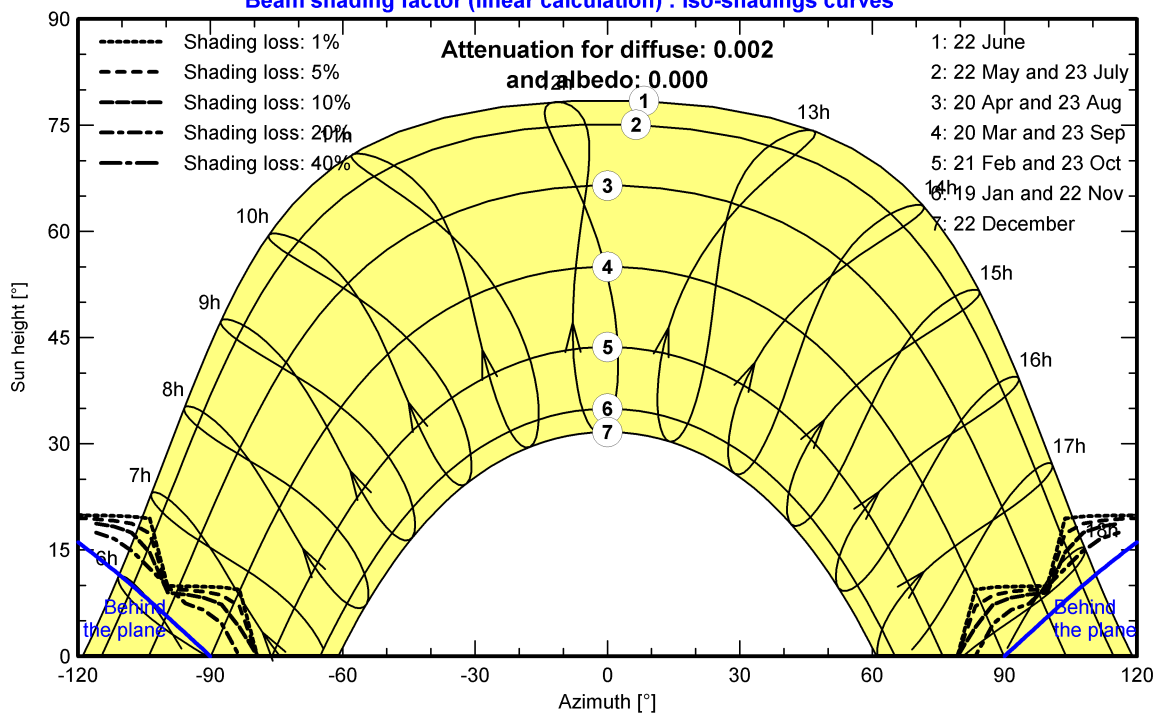
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 30°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves





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Main results

Simulation step

Hourly

System Production

Produced Energy

5605.6 kWh/year

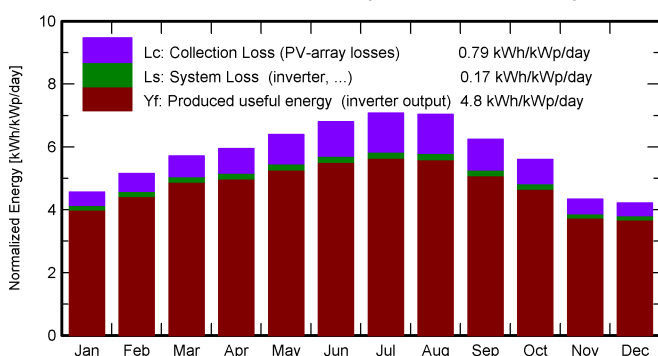
Specific production

1752 kWh/kWp/year

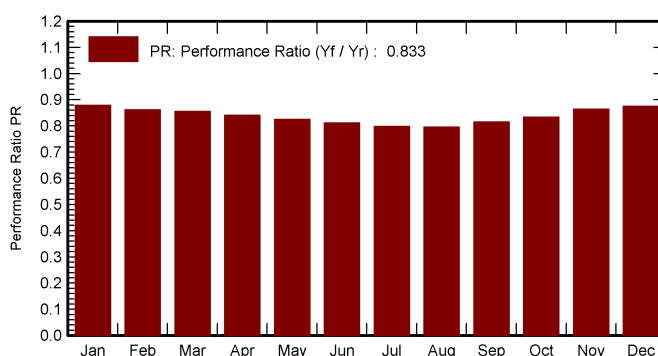
Perf. Ratio PR

83.30 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	89.6	31.05	10.25	141.3	139.1	411.4	397.3	0.878
February	104.2	34.44	11.33	144.2	141.8	411.8	397.3	0.861
March	147.2	59.20	13.70	177.0	173.3	502.1	484.8	0.856
April	167.8	69.63	16.06	178.3	174.3	497.0	479.4	0.840
May	206.7	73.53	19.63	198.3	193.2	542.8	523.6	0.825
June	222.4	73.93	24.07	204.0	198.6	548.6	529.5	0.811
July	232.4	75.66	27.46	219.5	214.4	580.3	560.2	0.797
August	212.5	68.47	27.52	218.2	213.5	575.7	555.7	0.796
September	163.5	60.22	23.64	187.4	183.3	505.9	488.6	0.815
October	131.5	44.32	19.98	173.7	170.8	479.8	463.1	0.833
November	88.7	36.91	14.44	130.3	128.1	372.9	360.1	0.864
December	79.4	26.75	11.36	130.7	128.6	379.2	366.0	0.875
Year	1846.1	654.12	18.33	2102.9	2058.9	5807.5	5605.6	0.833

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

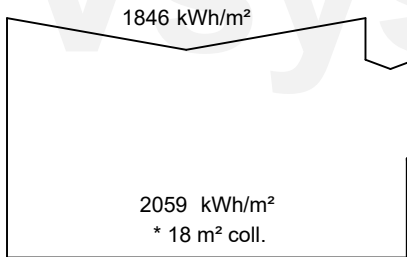
PR Performance Ratio



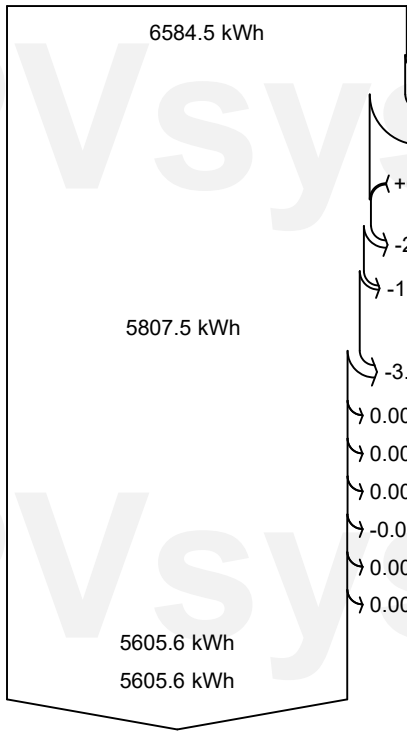
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Loss diagram



efficiency at STC = 17.83%



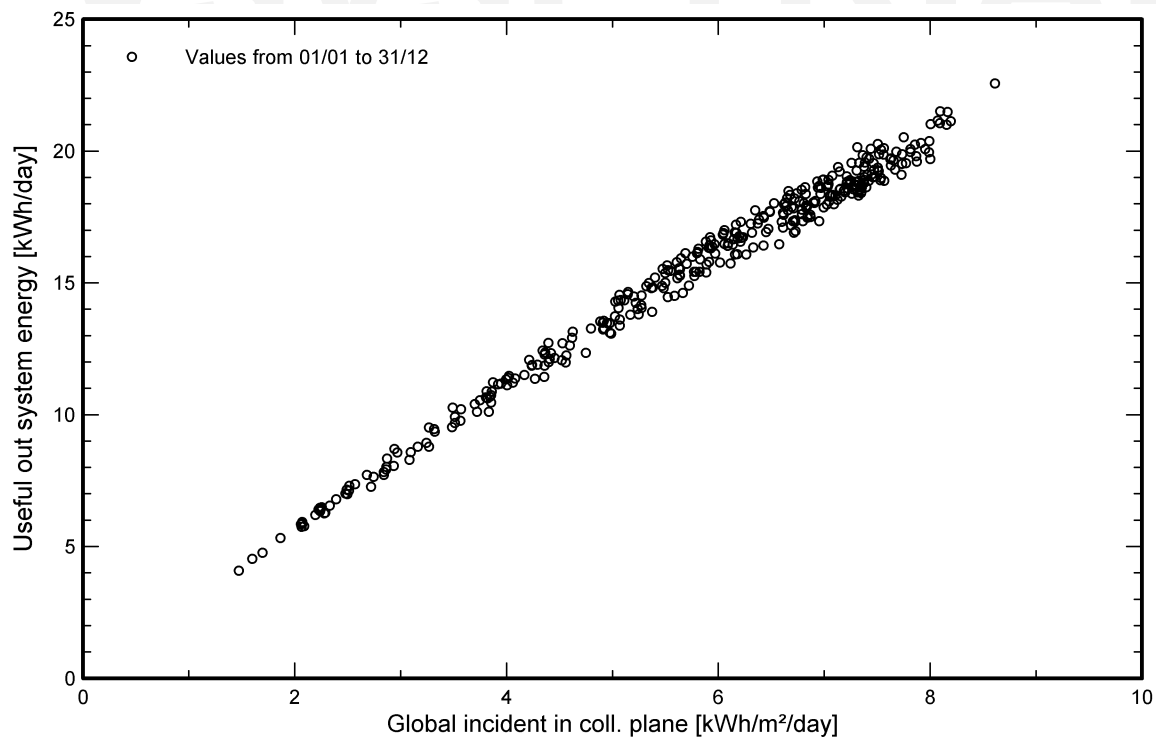
- Global horizontal irradiation
- Global incident in coll. plane
- Near Shadings: irradiance loss
- IAM factor on global
- Effective irradiation on collectors
- PV conversion
- Array nominal energy (at STC effic.)
- PV loss due to irradiance level
- PV loss due to temperature
- Module quality loss
- Module array mismatch loss
- Ohmic wiring loss
- Array virtual energy at MPP
- Inverter Loss during operation (efficiency)
- Inverter Loss over nominal inv. power
- Inverter Loss due to max. input current
- Inverter Loss over nominal inv. voltage
- Inverter Loss due to power threshold
- Inverter Loss due to voltage threshold
- Night consumption
- Available Energy at Inverter Output
- Energy injected into grid



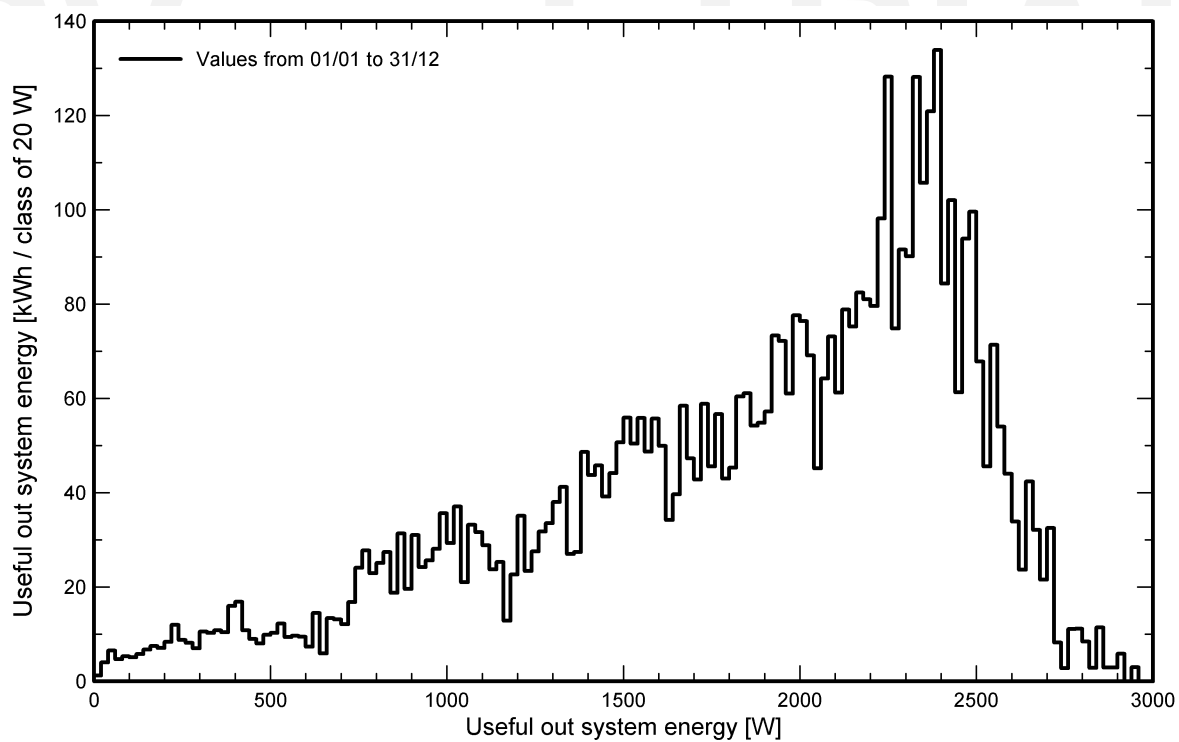
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

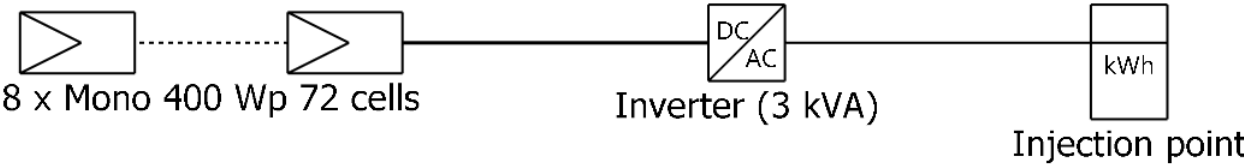




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Single-line diagram



PV module	Mono 400 Wp 72 cells
Inverter	3 kWac inverter
String	8 x Mono 400 Wp 72 cells

Fazouane

VC0 : Fazouane

30/09/26